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C:\WINDOWS\system32\cmd. x Windows PowerShell x Windows PowerShell x Windows PowerShell x + v - o x

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry>notepad mlflow_model_registry.py

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry>.\venv\Scripts\python.exe mlflow_model_registry.py
=====
W11D2 - MLFLOW MODEL REGISTRY & VERSIONING
=====

Dataset: Iris
Training samples: 120
Testing samples : 30

=====
RUNNING 5 MLFLOW EXPERIMENTS
=====

Experiment 1
Run ID : 68b98959547146eab89b29665a47b2e7
Parameters: {'n_estimators': 50, 'max_depth': 2, 'min_samples_split': 2}
Accuracy : 0.9333

Experiment 2
Run ID : 931c20408a364750ba2abbb81907156f
Parameters: {'n_estimators': 100, 'max_depth': 3, 'min_samples_split': 2}
Accuracy : 0.9667

Experiment 3
Run ID : 02586ae4500648b08ac4ee52feacef
Parameters: {'n_estimators': 150, 'max_depth': 4, 'min_samples_split': 2}
Accuracy : 0.9667

Experiment 4
Run ID : 60a1ae37afe9440e94084a316572996d
Parameters: {'n_estimators': 200, 'max_depth': 5, 'min_samples_split': 2}
Accuracy : 0.9000
```

```
Experiment 5
Run ID : ab00418e2ff04dbdad46dc71aa5aba6
Parameters: {'n_estimators': 250, 'max_depth': None, 'min_samples_split': 2}
Accuracy : 0.9000

=====
EXPERIMENT COMPARISON
=====
Experiment 1 | Accuracy: 0.9333 | Parameters: {'n_estimators': 50, 'max_depth': 2, 'min_samples_split': 2}
Experiment 2 | Accuracy: 0.9667 | Parameters: {'n_estimators': 100, 'max_depth': 3, 'min_samples_split': 2}
Experiment 3 | Accuracy: 0.9667 | Parameters: {'n_estimators': 150, 'max_depth': 4, 'min_samples_split': 2}
Experiment 4 | Accuracy: 0.9000 | Parameters: {'n_estimators': 200, 'max_depth': 5, 'min_samples_split': 2}
Experiment 5 | Accuracy: 0.9000 | Parameters: {'n_estimators': 250, 'max_depth': None, 'min_samples_split': 2}

=====
BEST EXPERIMENT
=====
Experiment : 2
Run ID : 931c20408a364750ba2abbb81907156f
Accuracy : 0.9667
Parameters: {'n_estimators': 100, 'max_depth': 3, 'min_samples_split': 2}

=====
REGISTERING BEST MODEL
=====
Successfully registered model 'Iris_RandomForest_Model'.
2026/09/16 20:40:21 WARNING mlflow.tracking._model_registry.fluent: Run with id 931c20408a364750ba2abbb81907156f has no artifacts at
artifact path 'model', registering model based on models:/m-a3f0d0909f0a48d2b08a526748d95bd5 instead
Created version '1' of model 'Iris_RandomForest_Model'.
Registered Model : Iris_RandomForest_Model
Model Version : 1

=====
LOADING REGISTERED MODEL
=====
Model URI : models:/Iris_RandomForest_Model/1
```

```
C:\WINDOWS\system32\cmd. x Windows PowerShell x Windows PowerShell x Windows PowerShell x + v - 0 x
Model Type : RandomForestClassifier

=====
REGISTERED MODEL PREDICTION TEST
=====
Test Predictions:
[0 2 1 1 0]

Actual Values:
[0 2 1 1 0]

=====
REGISTERED MODEL EVALUATION
=====
Registered Model Accuracy: 0.9667

Classification Report:
precision    recall  f1-score   support

   setosa    1.00    1.00    1.00     10
  versicolor 1.00    0.90    0.95     10
   virginica 0.91    1.00    0.95     10

 accuracy    0.97    0.97    0.97     30
  macro avg   0.97    0.97    0.97     30
weighted avg   0.97    0.97    0.97     30

=====
W11D2 TASK COMPLETED
=====
MLflow Tracking URI : sqlite:///mlflow.db
Experiment           : W11D2_Iris_Model_Registry
Registered Model     : Iris_RandomForest_Model
Registered Version    : 1
Best Accuracy        : 0.9667

=====
```

```
precision    recall  f1-score   support

   setosa    1.00    1.00    1.00     10
  versicolor 1.00    0.90    0.95     10
   virginica 0.91    1.00    0.95     10

 accuracy    0.97    0.97    0.97     30
  macro avg   0.97    0.97    0.97     30
weighted avg   0.97    0.97    0.97     30

=====
W11D2 TASK COMPLETED
=====
MLflow Tracking URI : sqlite:///mlflow.db
Experiment           : W11D2_Iris_Model_Registry
Registered Model     : Iris_RandomForest_Model
Registered Version    : 1
Best Accuracy        : 0.9667

=====
MLFLOW UI COMMAND
=====
mlflow server --backend-store-uri sqlite:///mlflow.db --host 127.0.0.1 --port 5000

=====
MODEL SERVING COMMAND
=====
mlflow models serve --model-uri "models:/Iris_RandomForest_Model/1" --port 5001 --env-manager local

=====
MODEL REGISTRY DETAILS
=====
Model Name      : Iris_RandomForest_Model
Model Version   : 1
Model URI       : models:/Iris_RandomForest_Model/1

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry>
```

W11D2: MLflow Model Registry x Complete MLflow Tasks x Home - MLflow x Ask Gemini

127.0.0.1:5000/?utm_source=chatgpt.com

Welcome to MLflow

Information about MLflow telemetry
MLflow collects usage data to improve the product. To confirm your preferences, please visit the settings page in the navigation sidebar. To learn more about what data is collected, please visit the [documentation](#).

Getting Started

New to MLflow?
Explore MLflow's core features with pre-populated sample data including traces, evaluations, and prompts.

Tracking
Capture and debug LLM interactions and agent workflows.

Evaluation
Measure and compare LLM quality with built-in and custom scores.

Prompts
Version control and manage prompts with aliases across teams.

AI Gateway
Unified API for multiple LLM providers with rate limiting.

Model Training
Track experiments with parameters, metrics, and artifacts.

Recent Experiments

Name	Time created	Last modified	Description	Tags
W11D2_ML_Model_Registry	09/16/2026 07:55:03 PM	09/16/2026 07:55:03 PM	-	
Default	09/16/2026 07:55:03 PM	09/16/2026 07:55:03 PM	-	

W11D2: MLflow Model Registry x Complete MLflow Tasks x Runs - Experiment 1 - MLflow x Ask Gemini

127.0.0.1:5000/?utm_source=chatgpt.com/#/experiments/1/runs?workflowType=genai&searchFilter=&orderByKey=attributes.start_time&orderBy...

Training runs

metrics: mse + 1 and param: model = 'tree'

State: Active

Run Name	Created	Dataset	Duration	Source	Models	Metrics
Experiment_1	21 minutes ago	-	24.7s	mlflow...	model	accuracy: 0.9
Experiment_1	21 minutes ago	-	24.6s	mlflow...	model	accuracy: 0.9
Experiment_1	22 minutes ago	-	23.8s	mlflow...	model	accuracy: 0.9000000...
Experiment_1	22 minutes ago	-	23.1s	mlflow...	ml_randomforest_model...	accuracy: 0.9000000...
Experiment_1	23 minutes ago	-	38.6s	mlflow...	model	accuracy: 0.9333333...

5 matching runs

W11D2: MLflow Model Registry x Complete MLflow Tasks x Run 68b98959547146eab89b29665a47b2e7

127.0.0.1:5000/?utm_source=chatgpt.com/#/experiments/1/runs/68b98959547146eab89b29665a47b2e7

mlflow 3.10.0

Model training

W11D2_ML_Model_Registry > Runs > Experiment_1

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (1)

Search metrics

Metric	Value	Model
accuracy	0.6133333333333333	model

Parameters (3)

Search parameters

Parameter	Value
n_estimators	50
max_depth	2
min_samples_split	2

Logged models (1)

Model attributes						No dataset	Parameters			
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	max_depth	min_samples_split	n_estimators
Output	0	model	Ready	23 minutes ago	-	-	0.6133333333333333	2	2	50

About this run

Created at 09/16/2026 08:38:06 PM

Created by Sergana CA

Experiment ID 1

Status Finished

Run ID 68b98959547146eab89b29665a47b2e7

Duration 38.6s

Source mlflow_model_registry

Registered prompts -

Datasets None

Tags Add tags

Registered models None

21:01 16-09-2026

W11D2: MLflow Model Registry x Complete MLflow Tasks x Run 931c20408a364750ba2abb81907156f

127.0.0.1:5000/?utm_source=chatgpt.com/#/experiments/1/runs/931c20408a364750ba2abb81907156f

mlflow 3.10.0

Model training

W11D2_ML_Model_Registry > Runs > Experiment_2

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (1)

Search metrics

Metric	Value	Model
accuracy	0.6666666666666667	model

Parameters (3)

Search parameters

Parameter	Value
n_estimators	100
max_depth	3
min_samples_split	2

Logged models (1)

Model attributes						No dataset	Parameters			
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	max_depth	min_samples_split	n_estimators
Output	0	model	Ready	23 minutes ago	931c20408a364750ba2abb81907156f	-	0.6666666666666667	3	2	100

About this run

Created at 09/16/2026 08:38:43 PM

Created by Sergana CA

Experiment ID 1

Status Finished

Run ID 931c20408a364750ba2abb81907156f

Duration 23.1s

Source mlflow_model_registry

Registered prompts -

Datasets None

Tags Add tags

Registered models 931c20408a364750ba2abb81907156f

21:01 16-09-2026

W11D2: MLflow Model Registry x Complete MLflow Tasks x Run 02586ae4500648b08ac4ee52fe1eacef

127.0.0.1:5000/?utm_source=chatgpt.com/#/experiments/1/runs/02586ae4500648b08ac4ee52fe1eacef

mlflow 3.10.0

Model training

W11D2_ML_Model_Registry > Runs > Experiment_3

Overview Model metrics System metrics Traces Artifacts

Description

No description

Metrics (1)

Search metrics

Metric	Value	Model
accuracy	0.9566666666666667	model

Parameters (3)

Search parameters

Parameter	Value
n_estimators	150
max_depth	4
min_samples_split	2

Logged models (1)

Model attributes				No dataset		Parameters				
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	max_depth	min_samples_split	n_estimators
Output	0	model	Ready	23 minutes ago	-	-	0.9566666666666667	4	2	150

About this run

Created at 09/16/2026 08:38:08 PM

Created by Sargana CA

Experiment ID 1

Status Finished

Run ID 02586ae4500648b08ac4ee52fe1eacef

Duration 23.8s

Source mlflow_model_registry

Registered prompts -

Datasets None

Tags

Add tags

Registered models None

21:02 16-09-2026

W11D2: MLflow Model Registry x Complete MLflow Tasks x Run 60a1ae37afe9440e94084a316572996d

127.0.0.1:5000/?utm_source=chatgpt.com/#/experiments/1/runs/60a1ae37afe9440e94084a316572996d

mlflow 3.10.0

Model training

W11D2_ML_Model_Registry > Runs > Experiment_4

Overview Model metrics System metrics Traces Artifacts

Description

No description

Metrics (1)

Search metrics

Metric	Value	Model
accuracy	0.9	model

Parameters (3)

Search parameters

Parameter	Value
n_estimators	200
max_depth	5
min_samples_split	2

Logged models (1)

Model attributes				No dataset		Parameters				
Type	Step	Model name	Status	Created	Registered models	Dataset	accuracy	max_depth	min_samples_split	n_estimators
Output	0	model	Ready	22 minutes ago	-	-	0.9	5	2	200

About this run

Created at 09/16/2026 08:38:32 PM

Created by Sargana CA

Experiment ID 1

Status Finished

Run ID 60a1ae37afe9440e94084a316572996d

Duration 24.4s

Source mlflow_model_registry

Registered prompts -

Datasets None

Tags

Add tags

Registered models None

21:02 16-09-2026

W11D2: MLflow Model Registry x Complete MLflow Tasks x Run ab00418e2ff04dbdac46dc71aa5aba6 x Ask Gemini

127.0.0.1:5000/?utm_source=chatgpt.com/experiments/1/runs/ab00418e2ff04dbdac46dc71aa5aba6

Experiment 5

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (1)

Metric	Value	Model
accuracy	0.9	model

Parameters (3)

Parameter	Value
n_estimators	250
max_depth	None
min_samples_split	2

Logged models (1)

Type	Step	Model name	Status	Created	Registered models	Dataset	No dataset	Parameters	max_depth	min_samples_split	n_estimators
Output	0	model	Ready	22 minutes ago	-	-	accuracy	0.9	None	2	250

About this run

Created at 09/16/2026 08:58 PM
 Created by Sanjana CA
 Experiment ID 1
 Status Finished
 Run ID ab00418e2ff04dbdac46dc71aa5aba6
 Duration 24.7s
 Source mlflow_model_registry
 Registered prompts --

Datasets None

Tags Add tags

Registered models None

```
C:\WINDOWS\system32\cmd.exe x Windows PowerShell x Windows PowerShell x Windows PowerShell x
ana%20CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry\mlruns\1\931c20408a364750ba2abbb81907156f/artifacts:
##### File model #####
No such artifact: 'model'
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry> $env:MLFLOW_TRACKING_URI="http://127.0.0.1:5000"
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry> echo $env:MLFLOW_TRACKING_URI
http://127.0.0.1:5000
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry> .\env\Scripts\mlflow.exe models serve --model-uri "models:/Iris_RandomForest_Model/1" --port 5001 --env-manager local
2026/09/16 20:52:55 INFO mlflow.models.flavor_backend_registry: Selected backend for flavor 'python_function'
2026/09/16 20:52:55 INFO mlflow.pyfunc.backend: === Running command 'uvicorn --host 127.0.0.1 --port 5001 --workers 1 mlflow.pyfunc.scoring_server.app:app'
2026/09/16 20:53:42 WARNING mlflow.utils.requirements_utils: Detected one or more mismatches between the model's dependencies and the current Python environment:
- numpy (current: 2.5.1, required: numpy==2.5.3)
- scikit-learn (current: 1.9.0, required: scikit-learn==1.9.1)
- scipy (current: 1.18.0, required: scipy==1.18.1)
To fix the mismatches, call 'mlflow.pyfunc.get_model_dependencies(model_uri)' to fetch the model's environment and install dependencies using the resulting environment file.
2026/09/16 20:53:42 WARNING mlflow.pyfunc: The version of Python that the model was saved in, 'Python 3.12.5', differs from the version of Python that is currently running, 'Python 3.14.0', and may be incompatible
C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator DecisionTreeClassifier from version 1.9.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator RandomForestClassifier from version 1.9.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
INFO: Started server process [14680]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:5001 (Press CTRL+C to quit)
INFO: 127.0.0.1:59028 - "POST /invocations HTTP/1.1" 200 OK
```

```
C:\WINDOWS\system32\cmd.exe X Windows PowerShell X Windows PowerShell X Windows PowerShell X + - Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA> cd "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry"
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry> $body = '{"dataframe_split":{"columns":["sepal length (cm)","sepal width (cm)","petal length (cm)","petal width (cm)"],"data":[[5.1,3.5,1.4,0.2]]}}'
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry> Invoke-RestMethod -Uri "http://127.0.0.1:5001/invocations" -Method Post -ContentType "application/json" -Body $body

predictions
-----
{0}

PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D2_MLflow_Model_Registry>
```